

新學堂

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DATA SET 9. Refer to Exercise 4 (Section 15.2). The data in this table are from a paired-difference experiment with $n = 7$ pairs of observations.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

- Use Wilcoxon's signed-rank test to determine whether there is a significant difference between the two populations.
- Compare the results of part a with the result you got in Exercise 4 (Section 15.2). Are they the same? Explain.

(a) H_0 : population distributions 1 and 2 are identical
 H_a : distribution difference between the two populations

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9

$$d \quad 0.1 \quad 0.7 \quad 0.3 \quad -0.1 \quad 0.5 \quad 0.2 \quad 0.5$$

$$|d| \quad 0.1 \quad 0.7 \quad 0.3 \quad 0.1 \quad 0.5 \quad 0.2 \quad 0.5$$

$$\text{rank} \quad 1.5 \quad 7 \quad 4 \quad 1.5 \quad 5.5 \quad 3 \quad 5.5$$

$$\text{sign} \quad + \quad + \quad + \quad - \quad + \quad + \quad +$$

$$T^+ = 1.5 + 7 + 4 + 5.5 + 3 + 5.5 = 26.5$$

$$T^- = 1.5$$

$$T^* = \min(T^+, T^-) = 1.5$$

$$T_7 = 2$$

$$\therefore T^* = 1.5 < T_7 = 2$$

$\therefore$ Reject H_0

(b) No, the results are not the same. The sign test do not reject H_0 , but Wilcoxon signed-rank test reject H_0 .

The Wilcoxon signed-rank test is more powerful because it uses both the signs and ranks of the difference.

DATA
SET

DS1511

11. Machine Breakdowns The number of machine breakdowns per month was recorded for 9 months on two identical machines, A and B, used to make wire rope:

Month	A	B	$d = A - B$	Rank
1	3	7	-4	-
2	14	12	2	+
3	7	9	-2	-
4	10	15	-5	-
5	9	12	-3	-
6	6	6	0	0
7	13	12	1	+
8	6	5	1	+
9	7	13	-6	-

- a. Do the data provide sufficient evidence to indicate a difference in the monthly breakdown rates for the two machines? Test by using a value of α near .05.
- b. Can you think of a reason the breakdown rates for the two machines might vary from month to month?

(a) H_0 : population distributions A and B are identical

H_a : distribution difference in the monthly breakdown rates for A and B

$$n = 8$$

$$T^+ = 3.5 + 1.5 + 1.5 = 6$$

$$T^- = 6 + 3.5 + 7 + 1.5 + 8 = 29.5$$

$$T_0 = 4$$

$$\therefore T^+ = 6 > T_0 = 4$$

$\therefore$ Do not reject H_0 . There is insufficient evidence to detect a difference in the two machines

(b) The monthly breakdown rates may vary because of changes in workload, maintenance, or operating conditions.

15. Images and Word Recall A psychology class performed an experiment to determine whether a recall score in which instructions to form images of 25 words were given differs from an initial recall score for which no imagery instructions were given. Twenty students participated in the experiment with the results listed in the table.

Student	With Imagery	Without Imagery	Student	With Imagery	Without Imagery
1	20	5	11	17	8
2	24	9	12	20	16
3	20	5	13	20	10
4	18	9	14	16	12
5	22	6	15	24	7
6	19	11	16	22	9
7	20	8	17	25	21
8	19	11	18	21	14
9	17	7	19	19	12
10	21	9	20	23	13

- What three testing procedures can be used to test for differences in the distribution of recall scores with and without imagery? What assumptions are required for the parametric procedure? Do these data satisfy these assumptions?
- Use both the sign test and the Wilcoxon signed-rank test to test for differences in the distributions of recall scores under these two conditions.
- Compare the results of the tests in part b. Are the conclusions the same? If not, why not?

(a) The three procedures are paired t-test, sign test and Wilcoxon signed-rank test.

The differences between the paired observations must be normally distributed.

Likely yes. The differences show no extreme outliers, but a formal normality test is required to confirm.

(b)

Student	With Imagery	Without Imagery	Sign	Student	With Imagery	Without Imagery	Sign
1	20	5	+	11	17	8	+
2	24	9	+	12	20	16	+
3	20	5	+	13	20	10	+
4	18	9	+	14	16	12	+
5	22	6	+	15	24	7	+
6	19	11	+	16	22	9	+
7	20	8	+	17	25	21	+
8	19	11	+	18	21	14	+
9	17	7	+	19	19	12	+
10	21	9	+	20	23	13	+

H_0 : There is no difference in the distributions of recall scores

H_a : There is a difference in the distributions

Sign Test

$$n = 20$$

$$p\text{-value} = 2P(X \geq 20) \\ = 0$$

$$\therefore p\text{-value} = 0 < 0.05$$

$\therefore$ Reject H_0

Wilcoxon Signed-Rank Test

$$n = 20$$

$$T^- = 0$$

$$T_{\alpha/2} = 52$$

$$\therefore T^- = 0 < T_{\alpha/2} = 52$$

$\therefore$ Reject H_0

(c) Yes, the conclusions are the same. Both tests reject H_0 .

This is because all 20 differences are positive, showing a very strong and consistent effect that both tests easily detect.

6. Legos® The time required for kindergarten children to assemble a specific Lego creation was measured for children who had been instructed for four different lengths of time. Five children were randomly assigned to each instructional group. The length of time (in minutes) to assemble the Lego creation was recorded for each child in the experiment.

Training Period (hours)				
.5	1.0	1.5	2.0	
8	9	4	4	
14	7	6	7	
9	5	7	5	
12	8	8	5	
10	9	6	4	

Use the Kruskal–Wallis H -Test to determine whether there is a difference in the distribution of times for the four different lengths of instructional time. Use $\alpha = .01$.

H_0 : The distributions of assembly times are the same across all 4 training periods

H_a : The distributions of assembly times are not the same

Training Period (hours)				
.5	1.0	1.5	2.0	
8 (13)	9 (16)	4 (2)	4 (2)	
14 (20)	7 (10)	6 (7.5)	7 (10)	
9 (16)	5 (6.5)	7 (10)	5 (6.5)	
12 (19)	8 (13)	8 (13)	5 (6.5)	
10 (18)	9 (16)	6 (7.5)	4 (6.2)	

$$T_1 = 86 \quad T_2 = 60 \quad T_3 = 40 \quad T_4 = 24$$

$$\begin{aligned}
 H &= \frac{12}{N(N+1)} \left(\sum \frac{T_i^2}{n_i} \right) - 3(N+1) \\
 &= \frac{12}{20(21)} \left(\frac{86^2}{5} + \frac{60^2}{5} + \frac{40^2}{5} + \frac{24^2}{5} \right) - 3(21) \\
 &= 12.27
 \end{aligned}$$

$$\chi^2_{3,0.01} = 11.345$$

$$\therefore H = 12.27 < \chi^2_{3,0.01} = 11.345$$

$\therefore$ Reject H_0 . There is sufficient evidence to conclude that the distributions of assembly times are different for the 4 training periods.

8. Heart Rate and Exercise In Exercise 18 (Section 11.3), we presented data on the heart rates for samples of 10 men randomly selected from each of four age groups. Each man walked a treadmill at a fixed grade for a period of 12 minutes, and the increase in heart rate (the difference before and after exercise) was recorded (in beats per minute). The data are shown in the table.

	10-19	20-39	40-59	60-69
29	24	37	28	
33	27	25	29	
26	33	22	34	
27	31	33	36	
39	21	28	21	
35	28	26	20	
33	24	30	25	
29	34	34	24	
36	21	27	33	
22	32	33	32	
Total	309	275	295	282

- a. Do the data present sufficient evidence to indicate differences in location for at least two of the four age groups? Test using the Kruskal-Wallis H -test with $\alpha = .01$.
- b. Find the approximate p -value for the test in part a.
- c. Since the F -test in Chapter 11 and the H -test are both tests to detect differences in location of the four heart-rate populations, how do the test results compare? If the ANOVA F -test produced the test statistic $F = 0.87$ with p -value = 0.468, compare this p -value with the p -value in part b and explain the implications of the comparison.

	10-19	20-39	40-59	60-69
29 (21)	24 (8)	37 (39)	28 (18)	
33 (19.5)	27 (15)	25 (10.5)	29 (14)	
26 (12.5)	33 (29.5)	22 (8.5)	34 (39)	
27 (11.5)	31 (24)	33 (28.5)	36 (37.5)	
39 (40)	21 (3)	28 (11)	21 (3)	
35 (35)	28 (18)	26 (10.5)	20 (1)	
33 (28.5)	24 (8)	30 (23)	25 (10.5)	
29 (20)	34 (34)	34 (34)	24 (8)	
36 (37.5)	21 (3)	27 (15)	33 (25.5)	
22 (15.5)	32 (29.5)	33 (29.5)	32 (25.5)	
Total	309	275	295	282

$$\begin{aligned} T_1 &= 247.5 & T_2 &= 168 & T_3 &= 216.5 & T_4 &= 188 \\ n_1 &= 10 & n_2 &= 10 & n_3 &= 10 & n_4 &= 10 \end{aligned}$$

1a) H_0 : The distributions of heart-rate increases are the same for all four age groups

H_a : At least one age group has a different distribution of heart rate increases

$$\begin{aligned} H &= \frac{12}{N(N+1)} \sum \frac{T_i^2}{n_i} - 3(n+1) \\ &= \frac{12}{40(41)} \left(\frac{247.5^2}{10} + \frac{168^2}{10} + \frac{216.5^2}{10} + \frac{188^2}{10} \right) - 3(41) \\ &= 2.63 \end{aligned}$$

$$\therefore H < \chi_{3,0.01}^2 = 11.345$$

$\therefore$ Do not reject H_0 . There is no evidence of a difference in location

1b) $H \approx 2.63$, $df = 3$

$$0.10 < p < 0.90$$

$$\therefore p > 0.01$$

$\therefore$ Do not reject H_0

1c) $P = 0.87$, p -value = 0.468

$$\therefore p = 0.468 > 0.01 \therefore \text{Do not reject } H_0$$

The results are similar because both the Kruskal-Wallis test and the ANOVA F -test do not reject H_0 . Both tests indicate that there is no significant difference among the four age groups.